Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Young, Laurence R. Sc.D.		
Project Title:	Neuro-Vestibular Aspects of Artificial Gravity Created by Short-Radius Centrifugation		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurovestibular Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lry@mit.edu	**Fax:**	FY 617-258-8111
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-7759
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Department of Aeronautics and Astronautics		
PI Address 2:	77 Massachusetts Avenue		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	8
Comments:	Deceased as of August 2021.		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	24
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paloski, William (NASA Johnson Space Center) Oman, Charles (Massachusetts Institute of Technology) Cohen, Bernard (Mount Sinai School of Medicine) Cohen, Malcolm (NASA Ames Research Center) Dai, Mingjia (Mount Sinai School of Medicine) DiZio, Paul (Brandeis University) Hecht, Heiko (Massachusetts Institute of Technology) Mast, Fred (Massachusetts Institute of Technology) Stone, Leland (NASA Ames Research Center) Welch, Robert (NASA Ames Research Center)		
Grant/Contract No.:	NCC 9-58-NA00202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project responds to the NSBRI Neurovestibular Adaptation Team Question "What are the effects of artificial gravity on human orientation and eye, head, and limb movements, and what are the pros and cons of various types of artificial gravity as countermeasures against the effects of micro-gravity on neurovestibular function? " It was thought until recently that AG could be implemented only by using large radius rotating spacecraft. We have further challenged the assumption that 4-6 rpm is a natural barrier beyond which adaptation becomes unattainable. Fast-spinning small rotational devices are a practical alternative. An integrated set of studies carried out at 3 universities treats the practical issue of adapting subjects to AG rapidly and with minimal motion sickness (MS) or motor disturbance. We have shown that adaptation to high rotation rates (23rpm) appears in physiological measures and in subjective responses (MS scores, illusory tilt and cross axis vestibular ocular reflex) when confining head-movements to one quadrant in a single plane (yaw head turns while supine). By the end of the second exposure a significant reduction in all three measures to about 1/3 of initial values is found and the adaptation is maintained over a five-day rest period. Further, adaptation generalizes quickly to the same head turns made while reversing the direction of centrifuge rotation. Likewise, pilot data suggest that adaptation to yaw head turns made in the right quadrant generalizes to the left quadrant within a few minutes. The changes due to adaptation are retained between exposures and are partially maintained over a five-day rest period. The three different measures adapt to Coriolis stimulation in different ways, and an augmented visual-vestibular conflict did not facilitate adaptation uniformly. Artificial Gravity (AG) especially when combined with exercise is an attractive antidote to some of the deleterious effects of long duration space flight. Since practical constraints forbid large radius spinning spacecraft or tethers, emphasis has shifted to short-radius centrifuges (SRC). This entails intermittent exposure to high rotation rates and the likelihood of motion sickness when out-of-plane head movements are made. The concern that high-speed rotation would be unacceptable for reasons of motion sickness was addressed and essentially resolved. The 21 published articles from this grant include documented, peer-reviewed papers explaining just how adaptation to AG is achieved and maintained.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Head movements in a moving or rotating environment, such as in boats, airplanes, and automobiles are often provocative. The ability to control susceptibility to motion sickness by controlling the central time constant of the vestibular system is a major advance and has broad application on Earth.
Task Progress:	We have shown that after a period of adaptation while rotating at 23 rpm, rapid head movements are tolerable for at least 75% of normal subjects. Most of the adaptation can be accomplished within two or three exposures of 20-30 minutes while rotating, spread out over 3 days. Tilt illusions and the vestibulo-ocular reflex following head turns during rotation depend upon the final orientation of the head relative to gravity. The visual surround during adaptation is relatively unrelated to MS and illusory motion, and only the VOR cross axis adaptation seems to require exposure in the light. Arm movements similarly adapt and show little or no long-term after-effects. Anti-MS medication (promethazine) does not seem to interfere with the adaptation process, although it was not effective in reducing MS during high speed CCS. The adaptation is general enough to carry across to a new environment, and even to rotation in the opposite direction, but may not transfer across planes of semicircular canal stimulation. Incremental adaptation, by gradually increasing the rotation speed, is effective but has not been extended all the way up to 23 rpm. This program was approved for funding in 2004 under a new 3-year grant from NSBRI: "Neurovestibular Aspects Of Short-Radius Artificial Gravity: Toward a Comprehensive Countermeasure".
Bibliography Type:	Description: (Last Updated: 01/21/2026)
Articles in Peer-reviewed Journals	Brown EL, Hecht H, Young LR. "Sensorimotor aspects of high-speed artificial gravity: I. Sensory conflict in vestibular adaptation." Journal of Vestibular Research. 2002-2003;12(5-6):271-82. PMID: 14501103. , Sep-2003
Articles in Peer-reviewed Journals	Cohen B, Dai M, Raphan T. "The critical role of velocity storage in production of motion sickness." Ann N Y Acad Sci. 2003 Oct;1004:359-76. PMID: 14662476. , Oct-2003
Articles in Peer-reviewed Journals	Hecht H, Brown EL, Young LR. "Adapting to artificial gravity (AG) at high rotational speeds." J Gravit Physiol. 2002 Jul;9(1):P1-5. PMID: 14703662. , Jan-2003
Articles in Peer-reviewed Journals	Hecht H, Kavelaars J, Cheung CC, Young LR. "Orientation illusions and heart-rate changes during short-radius centrifugation." J Vestib Res. 2001;11(2):115-27. PMID: 11847455. , Jan-2003
Articles in Peer-reviewed Journals	Mast FW, Newby NJ, Young LR. "Sensorimotor aspects of high-speed artificial gravity: II. The effect of head position on illusory self motion." J Vestib Res. 2002-2003;12(5-6):283-9. PMID: 14501104. , Sep-2003
Articles in Peer-reviewed Journals	Meliga P, Hecht H, Young LR, Mast FW. "Artificial gravity--head movements during short-radius centrifugation: influence of cognitive effects." Acta Astronaut. 2005 May-Jun;56(9-12):859-66. PMID: 15835032. , Jun-2005
Articles in Peer-reviewed Journals	Soeda K, DiZio P, Lackner JR. "Balance in a rotating artificial gravity environment." Exp Brain Res. 2003 Jan;148(2):266-71. PMID: 12520417. , Jan-2003
Articles in Peer-reviewed Journals	Young LR. "Models for neurovestibular adaptation." J Vestib Res. 2003;13(4-6):297-307. Review. PMID: 15096673. , Oct-2003

Articles in Peer-reviewed Journals	Young LR, Hecht H, Lyne LE, Sienko KH, Cheung CC, Kavelaars J. "Artificial gravity: head movements during short-radius centrifugation." Acta Astronautica Volume 49, Issues 3–10, August–November 2001, 215-26. https://doi.org/10.1016/S0094-5765(01)00100-X , Jan-2004
Articles in Peer-reviewed Journals	Young LR, Sienko KH, Lyne LE, Hecht H, Natapoff A. "Adaptation of the vestibulo-ocular reflex, subjective tilt, and motion sickness to head movements during short-radius centrifugation." J Vestib Res. 2003;13(2-3):65-77. PMID: 14757910. , Oct-2003
Articles in Peer-reviewed Journals	Dai M, Kunin M, Raphan T, Cohen B. "The relation of motion sickness to the spatial-temporal properties of velocity storage." Exp Brain Res. 2003 Jul;151(2):173-89. PMID: 12783152. , Jul-2003
Articles in Peer-reviewed Journals	DiZio P, Lackner JR. "Sensorimotor aspects of high-speed artificial gravity: III. Sensorimotor adaptation." J Vestib Res. 2002-2003;12(5-6):291-9. Review. PMID: 14501105. , Sep-2003
Awards	Young LA. "Professor Laurence A. Young was the subject of a 'Festschrift' in his honor, a celebration and symposium recognizing his career in space research: the 6th Symposium on the Vestibular Organs and the Exploration of Space, held in Portland OR, October 2002." Jan-2003
Books/Book Chapters	Cohen B, Raphan T. "The physiology of the vestibulo-ocular reflex (VOR) Springer Handbook of Auditory Research." Ed. Highstein S, Popper A , Fay RR. Springer Verlag, Jan-2004
Books/Book Chapters	Young LR. "Artificial Gravity." in "Encyclopedia of Space Science and Technology." Ed. H. Mark. New York : John Wiley & Sons, Inc., 2003., Mar-2003
Books/Book Chapters	Young LR. "Spatial Orientation." in "Principles and practice of aviation psychology." Ed. PS Tsang, MA Vidulich. Mahwah, NJ : Lawrence Erlbaum, 2003., Mar-2003
Books/Book Chapters	Young LR, Henn V, Scherberger H. "Fundamentals of the theory of movement perception. Translation of Mach, E. (1875). Grundlinien der Lehre von den Bewegungsempfindungen, Leipzig: Wilhelm Engelmann." Kluwer Academic/Plenum Publishers., Jan-2004
Presentation	10. Young, L. R. "Models for Neurovestibular Adaptation " Oct-2002
Presentation	Brown, E. "The role of visual inputs in adaptation to short-radius centrifugation " Oct-2002
Presentation	Dai, M., M., Raphan, T., and Cohen, B. "Adaptation to vertiginous vestibular stimulation " Oct-2002
Presentation	Di Zio, P., and Lackner, J. R. "The effect of rotation speed on motor adaptation to artificial gravity " Apr-2002
Presentation	Hecht, H. "Sensorische und motorische Aspekte der kuenstlichen Schwerkraft " Jul-2001
Presentation	Hecht, H. "The role of sensory conflict for adaptation to AG " Apr-2002
Presentation	Hecht, H. & Young, L. R., et al. "Neurovestibular aspects of artificial gravity " Jan-2002
Presentation	Hecht, H. and Young, L. R. "Neurovestibular aspects of artificial gravity " Jan-2001
Presentation	Hecht, H. and Young, L. R. "Vestibular adaptation in rotating environments " Jan-2001
Presentation	Lackner, J. R., and Di Zio, P. "Adaptation to artificial gravity " Oct-2002
Presentation	Mast, F. "Head orientation and vestibular illusions in artificial gravity " Apr-2002
Presentation	Moore, S. T., Clement, G., Dai, M., Raphan, T., and Cohen, B. "Spatial orientation of the vestibulo-ocular reflex (VOR) in microgravity: Results from the Neurolab STS-90 and Cosmos 2044 and 2229 missions " Oct-2002
Presentation	Young, L. R. "Adaptation to vertiginous vestibular stimulation " Oct-2002
Presentation	Young, L. R. "Micro-gravity and artificial gravity: Two challenges to neuro-vestibular adaptation " Oct-2002
Presentation	Young, L. R. "Neurovestibular aspects of artificial gravity " Apr-2002
Presentation	Young, L. R., and Hecht, H. "Context-specific dual adaptation in the vestibulo-ocular system " Oct-2002
Presentation	Young, L. R., and Hecht, H. "Neurovestibular aspects of artificial gravity " May-2002
Presentation	Young, L. R., Hecht, H. and Brown, E. "Adapting to artificial gravity (AG) at high rotational speeds " Jun-2002
Presentation	Young, L. R., Hecht, H. and Brown, E. "Neurovestibular aspects of artificial gravity " May-2002
Presentation	Young, L. R., Oman1, C. M., Jarchow, T., Hecht, H., Cohen, B., Dai, M., Lackner, J., Di Zio, P. Mast, F. Cohen, M. M., Paloski, W. H., Stone, L. & Welch, R. B. "Neurovestibular aspects of artificial gravity created by short-radius centrifugation " Jan-2003
Presentation	Young, L. R., Sienko, K. H., Lyne, L. E., and Hecht, H. "Adaptation of the vestibulo-ocular reflex, subjective tilt and motion sickness to head movements during short radius centrifugation " Sep-2002